The Ranking Transfers, the Threshold Does Not: Early Dropout Prediction in Unseen Course Contexts

Doudou Coly and El Hadji Mamadou Nguer

***Abstract*—Early-warning systems for online higher education are usually trained and evaluated within a single institution, on random splits that place the same courses in training and test sets, and often with features computed over the whole course. Before such a system is deployed in a new programme or institution, two questions must be answered: how much predictive signal is available early in a course, and what survives the move to a course the model has never seen? We study both on the Open University Learning Analytics Dataset (OULAD; 32,593 enrolments, 7 modules), using only data from the first 10–50% of each presentation and a leave-one-module-out (LOMO) protocol. An audit of the evaluation pipeline first shows that three common choices—keeping students who have already withdrawn at the prediction date, using scores submitted after it, and comparing a pooled AUC with a per-module average—more than double the apparent transfer gap (0.033 versus 0.013). We then separate ranking from thresholding. Ranking transfers almost entirely (mean AUC cost 0.013 at 25% of the course), but the decision threshold does not: in one module the share of flagged students rises from 44% to 95% for an observed risk of 37%. Signed SHAP attributions show that the transferred model uses the same features as local models but reads them on absolute scales—engagement volumes, formative scores—whose meaning depends on course design. Re-expressing engagement and early scores as within-cohort percentile ranks, which needs no labels and is available at prediction time, cuts the mean flag-rate drift from 0.137 to 0.021 and raises LOMO AUC from 0.787 to 0.809, outperforming Platt recalibration on 100 labelled students. Finally, threshold drift turns into group disparity through course composition: a female–male false-positive gap of +0.188 appears under LOMO although no module shows one internally; relative features remove it.**

***Index Terms*—Learning analytics, dropout prediction, early warning systems, transferability, dataset shift, calibration, SHAP, algorithmic fairness, OULAD.**

I. Introduction

DROPOUT remains one of the main obstacles to the promise of online higher education. Learning management systems (LMS) record student activity from the first day of a course, which has made data-driven early-warning systems an attractive way to target support while there is still time to act. The Open University Learning Analytics Dataset (OULAD) [1] has become the reference benchmark for this task: it is open, large, and combines demographics, registration, virtual learning environment (VLE) clickstreams and assessments for 22 module presentations. A recent systematic review counts a steady stream of predictive studies built on it [2].

High accuracy on OULAD is no longer informative in itself. Two limitations recur in the recent literature and both matter when a model is to be used elsewhere. First, many studies compute predictors over the whole presentation; a 2026 regional fairness audit notes that its design is retrospective and that several of its predictors span the whole presentation [3]. Such models cannot inform intervention, which must happen early. Second, models are almost always evaluated inside one institution, on splits that keep the same courses on both sides. A 2026 study predicting learner outcomes from the first 25% of each presentation names its single-institution scope as a limit to external generalization [4]; a calibrated early-warning study on OULAD splits by course presentation, so other presentations of a test module can still appear in training [5]. Outside OULAD, the evidence that predictive models are course-specific is strong [6]–[8], yet the mechanism of the failure is rarely examined.

This gap is concrete for institutions that would deploy an early-warning system without a labelled history of their own, such as the authors' fully online university in Senegal, where the target courses differ from any training course in design, assessment calendar and population. The relevant question is not whether a model achieves a high AUC on held-out students of known courses, but what part of its behaviour survives the move to a course it has never seen, why, and what can be done before labels exist.

We address this question on OULAD with an evaluation that is honest about timing and context. Predictions use only data available at a cut-off day placed at 10–50% of each presentation, and we compare a random split grouped by student, protocol (A), with leave-one-module-out, protocol (B), in which each module plays the role of an unseen context. Our contributions are:

1) *A methodological audit.* We show that three choices common in OULAD pipelines inflate early performance and the apparent transfer gap: keeping students already withdrawn at the cut-off (19.5% of enrolments at 25%), using scores submitted after the cut-off, and comparing a pooled AUC in (A) with a per-module mean in (B). Corrected, the transfer gap falls from 0.033 to 0.013 (Table I).
2) *A decomposition of transfer.* We separate intrinsic module difficulty, the transfer cost on ranking, and the drift of the decision threshold. Ranking transfers almost entirely; the threshold does not, and the modules that look hardest are difficult rather than untransferable (Section IV-C).
3) *A mechanism.* Using a transferred-versus-local SHAP comparison and a decomposition of the drift into signed attributions, we show that the transferred model relies on the same features as local ones but reads them on absolute scales whose meaning depends on course design. Global

D. Coly and E. H. M. Nguer are with the Université Numérique Cheikh Hamidou Kane (UN-CHK), Dakar, Senegal.

|SHAP| importance cannot reveal this (Section IV-E).

4) *A label-free remedy.* Re-expressing engagement and early scores as percentile ranks within the cohort removes the threshold drift and improves ranking, and compares favourably with Platt recalibration on small labelled samples (Section IV-F).
5) *A fairness consequence.* We show that threshold drift across contexts converts into a gender disparity through course composition, an effect that pooled fairness audits read as discrimination and within-context audits do not (Section IV-G).

II. Related Work

A. Early prediction on OULAD

OULAD was released to support the at-risk prediction work of the Open University [1]. Studies have since varied the observation window, for instance predicting at several percentages of the course length [9], or from the first 25% of each presentation with deep architectures [4]. Wu et al. [5] use a 28-day window, calibrate a gradient-boosting model, and audit subgroup fairness and alert capacity; their split holds out course presentations, not modules. Hlosta et al. [10] address a related cold-start problem at the Open University by training on the running presentation itself rather than on legacy data. We complement these studies by holding out entire modules and by asking which parts of the model's behaviour survive.

B. Portability of predictive models across courses

The Open Academic Analytics Initiative showed that a model trained at one institution could be ported to others with moderate loss [11]. Course-level analyses are less optimistic: Gašević et al. [6] found that the predictive value of LMS indicators depends on instructional conditions, Conijn et al. [7] that models varied substantially across 17 courses of a single institution, and López-Zambrano et al. [8] that direct transfer between Moodle courses works only under some conditions. Riestra-González et al. [12] built course-agnostic early predictors from massive LMS logs, and Boyer and Veeramachaneni [13] studied transfer learning across MOOC offerings. These works measure whether accuracy transfers; we decompose *what* fails to transfer and why.

C. Dataset shift and calibration

A change of context is a case of dataset shift, in which covariate, prior or concept changes can affect a classifier [14]. In clinical prediction, it is well established that discrimination often survives a change of setting while calibration does not, and that simple updating of the intercept or slope on local data restores much of it [15], [16]. Platt scaling [17] is the corresponding tool for machine-learning classifiers. Learning analytics has paid less attention to calibration under transfer, although decisions to contact students are taken at a threshold.

D. Fairness of predictive student models

Fairness audits of student models compare error rates and alert rates across demographic groups [18], [19]. On OULAD, recent studies document accuracy–fairness trade-offs [20], regional disparities in recall [3] and policy-dependent alert-rate gaps [5]. When base rates differ between groups, error-rate parity and calibration cannot all hold [21], [22]; equal opportunity [23] focuses on true-positive rates. We add a structural observation: aggregating over contexts can create group disparities that exist in no context, an instance of Simpson's paradox [24].

III. Data and Methods

A. Dataset and target

OULAD [1] (CC-BY 4.0) contains 32,593 enrolments in 22 presentations of 7 modules (AAA–GGG) at the Open University, UK. Each enrolment has a final result among Distinction, Pass, Fail and Withdrawn. Our main target is $y = 1$ for Withdrawn or Fail (52.8% of enrolments), since both outcomes call for the same early support; Section IV-H repeats the analyses with Withdrawn only.

B. Early window, population and features

For a presentation p of length L_p days, the cut-off day is $d_c = \mathrm{round}(c\,L_p)$ with $c \in \{0.10, 0.20, 0.25, 0.33, 0.50\}$; $c = 0.25$ is the reference. The population at d_c consists of the students registered on or before d_c and not unregistered by d_c. Students who have already left cannot be helped by an early warning, and keeping them makes the task artificially easy (Section IV-A).

For each enrolment we compute eighteen predictors from data dated on or before d_c: *demographics* (gender, region, highest education, IMD band, age band, previous attempts, studied credits, disability); *registration* (registration date relative to the course start and days registered in advance); *VLE engagement* (total clicks, active days, distinct resources, first and last active day, clicks per active day); *early assessments* (number of assessments, and mean score of those, whose deadline *and* submission both fall on or before d_c). Missing engagement is coded as zero activity, missing first and last active days by out-of-range sentinels, and a missing early score as -1.

Leakage control: Beyond excluding the final result and the unregistration date from the predictors [25], we test the whole pipeline by invariance: removing every raw record dated after d_c (clicks, submissions, deadlines, unregistrations) must leave the feature table unchanged. The test detects, for instance, a filter on assessment deadlines alone, which lets late submissions through (4.0% of the retained submissions at $c = 0.25$).

C. Absolute and context-relative features

The *absolute* feature set uses the values above, as is usual. The *relative* set replaces the eight engagement and early-assessment features by their percentile rank within the presentation,

$$\tilde{x}_{ij} = \frac{1}{n_p}\Big(L_{ij} + \frac{E_{ij} + 1}{2}\Big), \tag{1}$$

where P_p is the population of presentation p at d_c, $n_p = |P_p|$, L_{ij} counts the students of P_p whose value of feature j is lower than x_{ij}, and E_{ij} those, i included, whose value equals it (average rank for ties). Demographic and registration features are unchanged. The transformation uses no labels and only data available at the cut-off, and it passes the invariance test.

D. Model and protocols

We use a random forest [26] of 300 trees with balanced class weights (scikit-learn [27], one-hot encoding of categorical variables, seed 0) and flag a student when the predicted probability $\hat{p} \geq 0.5$. Protocol (A) is a five-fold split stratified on y and grouped by student, so that no student appears in both training and test folds; its out-of-fold predictions cover every enrolment. Protocol (B) trains on six modules and predicts the seventh, for each module in turn.

Both protocols are summarized in the same way: metrics are computed on the students of each module, then averaged over modules. An AUC pooled over all modules in (A) would reward the model for separating modules with different base rates, which (B) cannot do by construction.

For each module m we define three quantities on the same students:

$$\text{difficulty:} \quad D_m = \mathrm{AUC}_A(m), \tag{2}$$

$$\text{ranking cost:} \quad C_m = \mathrm{AUC}_A(m) - \mathrm{AUC}_B(m), \tag{3}$$

$$\text{flag-rate drift:} \quad \Delta_m = \bar{f}_B(m) - \bar{f}_A(m), \tag{4}$$

where $\bar{f}(m)$ is the share of the module's students flagged. D_m measures how predictable a module is when the model has seen it, C_m how much ranking is lost when it has not, and Δ_m how far the volume of alerts moves.

E. Remedies

We compare two ways of correcting the transferred model. *Platt recalibration* [17] fits a logistic regression of y on $\mathrm{logit}(\hat{p}_B)$, with balanced weights as in the source model, on labelled students of the target module: either $k \in \{25, 50, 100, 200\}$ students drawn at random (50 draws, evaluation on the remaining students), or the first presentation of the module as a pilot cohort, evaluated on later presentations. *Relative features* (Eq. (1)) require no labels.

F. Context descriptors

To explain heterogeneity, each module and presentation is described by its assessment structure (number of assessments due before the cut-off, first non-exam deadline as a fraction of the presentation, share of the exam in the total weight), observable engagement (share of students with an early score, median clicks, number of VLE activity types) and population (size, at-risk rate, share of Withdrawn among at-risk students, share without A-levels, share aged 35 or more), plus the standardized distance of the assessment structure to that of the other modules. We report Spearman correlations with D_m, C_m and Δ_m at the module level ($n = 7$) and the presentation level ($n = 22$; presentations of a module share its LOMO model and are not independent), with Benjamini–Hochberg control of the false discovery rate within level [28].

G. Interpretability

We compute SHAP values [29] with TreeSHAP [30] for two models per module m: the *transferred* LOMO model, and a *local* model trained on m alone with two-fold cross-fitting grouped by student (each half is explained by the model trained on the other). Attributions of one-hot columns are summed back to their original variable. Exact TreeSHAP on unpruned forests is slow (about 1.7 s per student for 300 trees); since the SHAP value of a forest is the average of those of its trees, we estimate it without bias from 100 trees drawn at random, which reproduces the exact importance ranking with Spearman $\rho = 0.985$. We explain 400 students per module.

Two analyses follow. The *agreement* between transferred and local models is the Spearman correlation of their global importances (mean $|\phi|$) over the 18 variables. The *drift decomposition* averages signed attributions of the transferred model over the students of m and over feature families G:

$$\bar{\phi}_{m,G} = \frac{1}{|S_m|} \sum_{i \in S_m} \sum_{j \in G} \phi_{ij}. \tag{5}$$

By local accuracy, the sum over all families equals the mean difference between the predicted risk of m's students and the average prediction on the training modules.

H. Fairness

We compare true-positive rate (TPR), false-positive rate (FPR), precision (PPV) and flag rate between women and men, the most and least deprived IMD quintiles (Q1: 0–20%; Q5: 80–100%), and students with and without a declared disability. Confidence intervals (95%) come from a cluster bootstrap over students with 1,000 resamples [31]; the same resamples are used for all configurations, which gives paired intervals for changes in gaps.

I. Implementation and reproducibility

All analyses run in Python 3.12 with scikit-learn 1.9.1, pandas 3.0.6 and shap 0.54.1, with fixed seeds. A single script regenerates every figure, table and number of this article from the raw OULAD files, and an automated test suite checks the absence of leakage.

IV. Results

A. Methodological audit

The reference pipeline we started from, which keeps all enrolments and compares a pooled AUC in (A) with a per-module mean in (B), gives AUC 0.882 in (A) and 0.848 in (B) at 25% of the course, a transfer gap of 0.033 (Table I). Filtering early scores on their submission date changes little. Restricting the population to students still enrolled at the cut-off has the largest effect: at 25%, 19.5% of enrolments have already ended, 62.5% of all Withdrawn outcomes among them, including 2,678 unregistrations before the course started. Their last active day makes them trivial to predict, and their share ranges from 5.4% (GGG) to 28.0% (CCC), which confounds any comparison between modules. Finally, comparing like with like halves the gap again, from 0.027 to 0.013. Together, these

TABLE I
METHODOLOGICAL AUDIT AT THE 25% CUT-OFF (RANDOM FOREST). EACH ROW ADDS ONE CORRECTION TO THE PREVIOUS ONE. AUC OF PROTOCOL (A), RANDOM SPLIT GROUPED BY STUDENT, AND (B), LEAVE-ONE-MODULE-OUT (LOMO), MEAN OVER HELD-OUT MODULES.

Evaluation setting	AUC (A)	AUC (B)	Gap
1) Reference protocol (all enrolments, pooled AUC in (A))	0.882	0.848	0.033
2) + early scores filtered on submission date	0.880	0.849	0.031
3) + students still enrolled at the cut-off only	0.815	0.787	0.027
4) + same aggregation in (A) and (B)	0.800	0.787	0.013
5) + context-relative features	0.816	0.809	0.008

choices make the apparent transfer gap more than twice its corrected value. Context-relative features reduce it further to 0.008.

B. How early can we predict?

Fig. 1 shows performance against the length of the observed window. With relative features, AUC rises from 0.749 at 10% of the course to 0.878 at 50% in (A), and from 0.740 to 0.871 in (B); the gap between protocols stays between 0.006 and 0.009 at every cut-off (0.011–0.015 with absolute features). With absolute features, the transferred model flags too many students: at 25%, its recall exceeds that of (A) (0.722 versus 0.649) while its precision falls (0.617 versus 0.669). With relative features, recall and precision of (B) track those of (A) (0.652 versus 0.648; 0.684 versus 0.697). Reporting recall alone, a common practice for at-risk prediction, would have made the transferred model look better than the local one.

C. The ranking transfers, the threshold does not

Table III and Fig. 2 separate the three quantities per module at 25%. The ranking cost is small: 0.013 on average with absolute features, and below 0.015 for six modules out of seven; BBB is the exception (0.046). Modules AAA and GGG, which a naive reading of LOMO results would call "poorly transferable", are in fact *difficult*: their AUC is already low when the model has seen them (0.733 and 0.739), and it drops by only 0.013 when it has not.

The threshold is another matter. In GGG, the transferred model flags 95% of students, against 44% for the model that has seen the module and an observed risk of 37%: recall climbs from 0.66 to 0.99 and precision falls from 0.55 to 0.38. BBB drifts too (+0.19), CCC and DDD slightly downwards. The mean absolute drift is 0.137. An institution that deployed such a model in a GGG-like course would contact almost every student.

D. Why GGG drifts

Table II shows how GGG differs from the other modules. Its first non-exam assessment falls at about a quarter of the presentation instead of 7–13%, its assignments are formative (the exam carries the whole weight), and only half of its students have an early score at the cut-off. Its students also use the VLE much less: their median click count before the cut-off is 81 against 432 in the other modules, the 11th percentile of the latter. Elsewhere, students with at most 81 clicks are at risk in 79% of cases; in GGG, 47%. Elsewhere, a student without an early score is at risk in 98% of cases; in GGG, where 47% of students have none yet, 51%. Early scores are also lower (median 69 against 79). The drift is not limited to the period before the first deadline: it reaches +0.60 at 33% and +0.52 at 50% of the course.

In short, the model has learned absolute thresholds—"few clicks" or "no score yet" means risk—whose meaning depends on the design of the course. Table IV gives the associations that survive multiple-testing control. Difficulty is associated with the share of older students (Spearman $\rho = -0.96$ over modules, -0.79 over presentations). Drift is larger where at-risk students fail rather than withdraw ($\rho = -0.71$), where students are older and less qualified, and where the VLE is used less. The structural distance of the assessment design to the other modules is associated with drift, not with the ranking cost, which is weakly associated with an earlier and denser assessment calendar. With seven modules, these associations are exploratory; the experiments below test the mechanism directly.

E. Interpretability: same features, wrong thresholds

If transfer failed because different features matter in different modules, global importances would differ between transferred and local models. They mostly do not. Local models rank the 18 variables similarly across modules (mean pairwise Spearman 0.81), with the mean early score first everywhere. Agreement between transferred and local importances is moderate with absolute features (mean 0.60, 3.3 of the top five variables in common) and higher with relative features (0.82, 4.0 in common), an improvement in 6 modules out of seven (Fig. 3a). Crucially, GGG—the module that drifts most—shows one of the *highest* agreements with absolute features (0.79). Its problem is not which features are used but how they are read.

The drift decomposition of Eq. (5) makes this visible (Fig. 3b). On GGG's students, the transferred absolute model raises predicted risk by +0.30 above its training average: +0.19 from early-score features and +0.12 from engagement, while demographics contribute -0.02. Across modules, this total follows the flag-rate drift ($\rho = 0.68$). With relative features the decomposition vanishes (-0.04 for GGG; Fig. 3c). Global importance, which discards the sign of attributions, cannot detect a threshold problem; signed attributions on the target population can.

F. Remedies

Table V compares the two corrections. Platt recalibration on 100 labelled students of the target module reduces the mean absolute drift from 0.137 to 0.058 and repairs GGG (drift -0.04, precision $0.38 \rightarrow 0.55$). Using the first presentation as a pilot cohort is less effective on average (0.096), because presentations of the same module also differ; it still helps GGG (precision 0.49, recall 0.75 instead of 0.99). Recalibration leaves

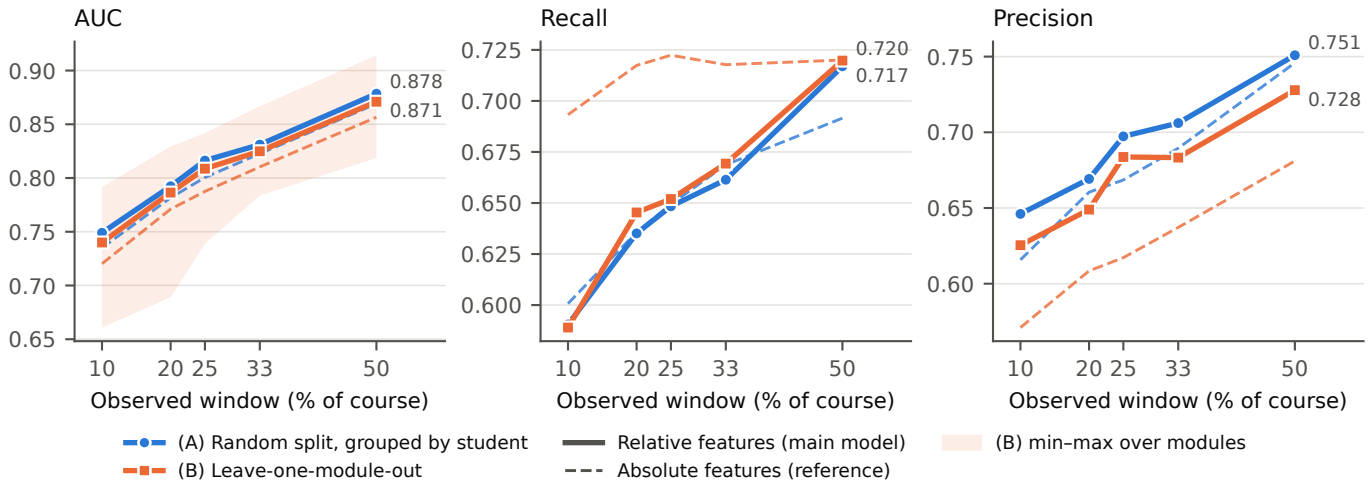


Fig. 1. Performance against the observed share of the presentation (students still enrolled at the cut-off). Solid lines: relative features; dashed lines: absolute features. Both protocols are averaged over modules in the same way; the band shows the range over held-out modules in (B) with relative features.

TABLE II
MODULE CHARACTERISTICS AT THE 25% CUT-OFF, STUDENTS STILL ENROLLED AT THE CUT-OFF. AT-RISK: WITHDRAWN OR FAIL. FIRST DEADLINE: FIRST NON-EXAM ASSESSMENT, AS A FRACTION OF THE PRESENTATION LENGTH. EXAM WEIGHT: SHARE OF THE EXAM IN THE TOTAL ASSESSMENT WEIGHT. EARLY SCORE: SHARE OF STUDENTS WITH AT LEAST ONE ASSESSMENT SUBMITTED BEFORE THE CUT-OFF. VALUES ARE AVERAGED OVER THE PRESENTATIONS OF EACH MODULE.

Module	n	At-risk (%)	Withdrawn among at-risk (%)	First deadline	Exam weight (%)	Early score (%)	Median clicks	Aged 35+ (%)
AAA	698	23.9	45.5	0.07	50	98.1	588	52.6
BBB	6,202	39.5	28.0	0.07	50	93.4	224	34.9
CCC	3,191	47.4	48.4	0.07	67	93.1	402	30.3
DDD	4,984	47.7	40.7	0.09	50	93.4	382	26.2
EEE	2,470	33.2	31.6	0.13	50	90.9	586	24.9
FFF	6,289	42.0	35.4	0.08	50	94.5	866	26.2
GGG	2,398	36.9	17.6	0.24	100	53.3	81	39.0

TABLE III
RANKING VERSUS THRESHOLD TRANSFER PER MODULE (25% CUT-OFF, STUDENTS STILL ENROLLED). AUC AND FLAG RATE ($p \geq 0.5$) ARE COMPUTED ON THE STUDENTS OF EACH MODULE; (A) USES OUT-OF-FOLD PREDICTIONS OF THE RANDOM SPLIT, (B) THE MODEL TRAINED ON THE SIX OTHER MODULES.

	Ranking (absolute features)			Flagged students, % (absolute)			Relative features, (B)	
Module	AUC (A)	AUC (B)	Cost	Observed risk	(A)	(B)	AUC	Flagged (%)
AAA	0.733	0.720	+0.013	24	35	44	0.793	37
BBB	0.796	0.750	+0.046	39	37	56	0.804	40
CCC	0.830	0.825	+0.004	47	44	40	0.813	45
DDD	0.837	0.830	+0.007	48	44	39	0.842	39
EEE	0.841	0.844	−0.004	33	26	33	0.838	30
FFF	0.828	0.816	+0.012	42	36	34	0.833	36
GGG	0.739	0.726	+0.013	37	44	95	0.739	30
Mean	0.800	0.787	+0.013	39	38	49	0.809	37

the ranking unchanged and moves F1 little (0.639 to 0.647 on average), since F1 rewards over-flagging when risk is common.

Relative features do better without any label: mean absolute drift 0.021, GGG drift −0.01, and an AUC of 0.809 on unseen modules—higher than that of the absolute model on modules it has seen (0.800). Mean precision and F1 of (B) rise to 0.684 and 0.661. The ranks discard information on the absolute level of engagement in a cohort, but this information was precisely what did not transfer.

G. Fairness

Table VI and Fig. 4 report error-rate gaps between groups. With absolute features, transfer creates a gender gap that does not exist when modules are seen: the FPR gap between women and men moves from −0.007 [−0.019, +0.005] in (A) to +0.188 [+0.174, +0.201] in (B). Yet inside every module the transferred model's FPR is *lower* for women (the largest within-module gap is −0.023; Fig. 4c). The pooled gap is a composition effect: women make up 88% of BBB and 81% of GGG, the two modules whose threshold drifts upwards, so the drift of

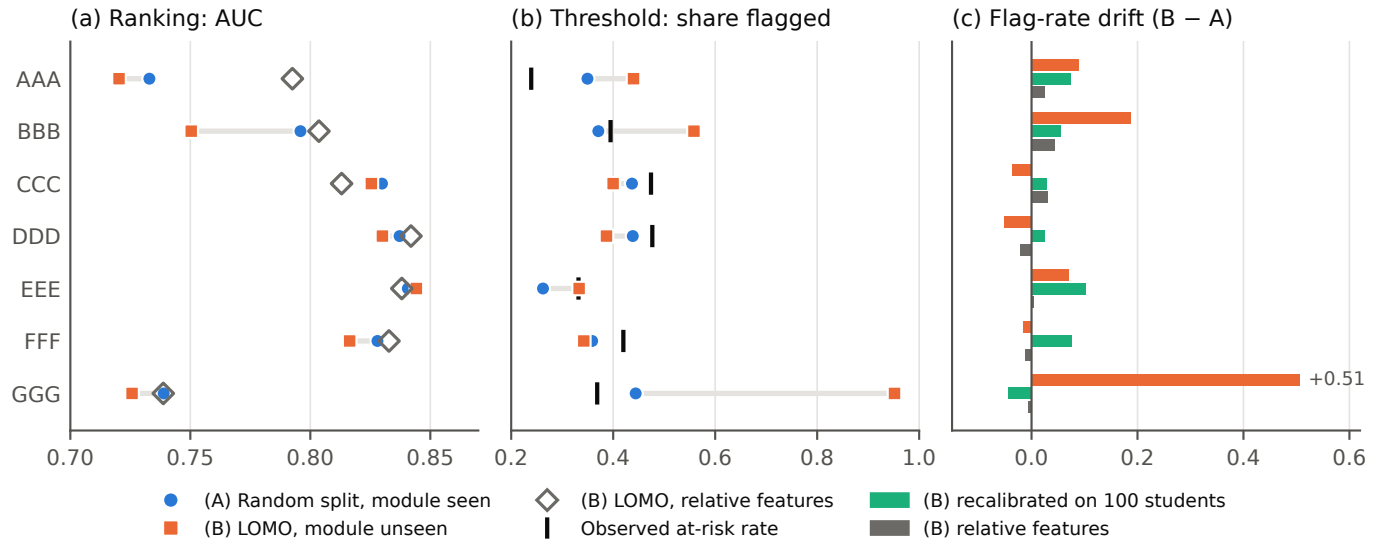


Fig. 2. Ranking versus threshold per module at the 25% cut-off. (a) AUC when the module is seen (A) or unseen (B), and (B) with relative features. (b) Share of students flagged in (A) and (B) against the observed at-risk rate. (c) Flag-rate drift of the raw transferred model, after Platt recalibration on 100 labelled students of the module, and with relative features.

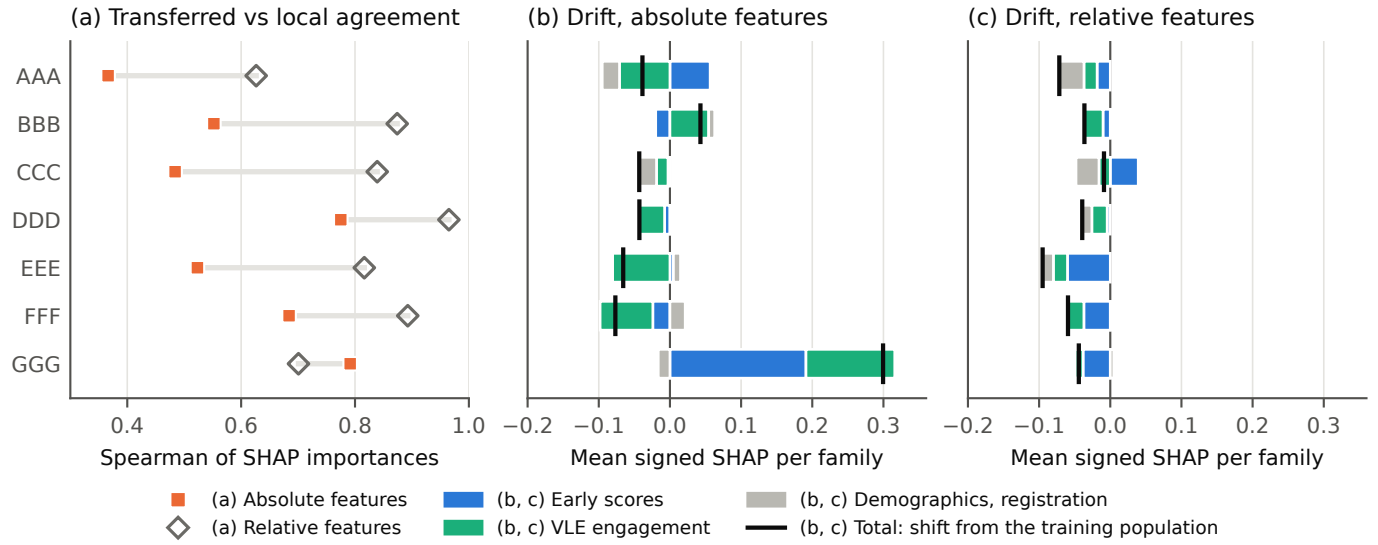


Fig. 3. Same features, wrong thresholds (25% cut-off). (a) Spearman correlation between the SHAP importances of the transferred model and of a model trained on the module itself. (b, c) Mean signed SHAP attribution of the transferred model on the students of each module, by feature family: the shift of their predicted risk from the training population.

these modules is attributed to gender when modules are pooled [24]. With relative features, the gap falls to +0.016 [+0.004, +0.027].

IMD and disability gaps behave differently. They exist under every protocol and reflect different base rates (51% at risk in Q1 against 33% in Q5): the most deprived students are flagged more often, correctly and incorrectly (FPR gap +0.148 in (A), +0.163 in (B)). Relative features reduce the FPR gap in (B) by −0.051 [−0.075, −0.026] without closing it, consistent with the impossibility of equalizing error rates and calibration when base rates differ [21], [22].

H. Robustness to the target definition

Restricted to students still enrolled at the cut-off, "Withdrawn only" becomes *future* withdrawal, a rarer (13.7% at 25%) and harder target (Table VII). At the 0.5 threshold, the balanced forest rarely flags such students (recall 0.13 in (B)), so we compare on AUC. The pattern holds: relative features lower the ranking cost from 0.021 to 0.006, the mean absolute drift from 0.052 to 0.028 (GGG: +0.18 to +0.03), and raise LOMO AUC from 0.630 to 0.668.

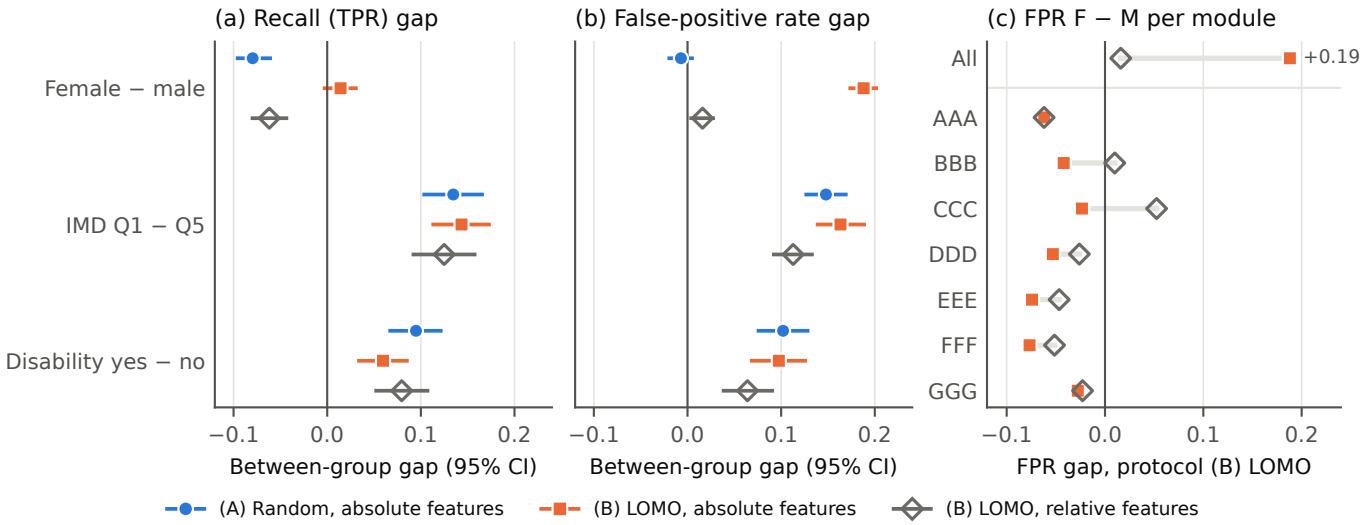


Fig. 4. Error-rate gaps between groups (25% cut-off, 95% cluster-bootstrap intervals). (a) Recall and (b) false-positive rate gaps. (c) Female–male FPR gap of the transferred model computed inside each module and over all modules: the pooled gap of the absolute model is a composition effect.

TABLE IV
CONTEXT DESCRIPTORS ASSOCIATED WITH THE THREE TRANSFER AXES (SPEARMAN'S ρ, BENJAMINI–HOCHBERG $q < 0.10$ WITHIN LEVEL, 69 TESTS IN TOTAL). MOD.: MODULES; PRES.: PRESENTATIONS, WHICH ARE NOT INDEPENDENT (THEY SHARE THE MODULE-LEVEL LOMO MODEL). DIFFICULTY: AUC (A); RANK. COST: AUC (A) − AUC (B); DRIFT: FLAG-RATE DRIFT. EXPLORATORY ASSOCIATIONS.

Level (n)	Axis	Descriptor	ρ	q
Mod. (7)	Difficulty	Share aged 35+	−0.96	0.016
Pres. (22)	Difficulty	Share aged 35+	−0.79	0.000
Pres. (22)	Difficulty	VLE activity types	+0.54	0.037
Pres. (22)	Difficulty	At-risk rate	+0.49	0.060
Pres. (22)	Rank. cost	First deadline	−0.55	0.037
Pres. (22)	Rank. cost	Share with early score	+0.52	0.042
Pres. (22)	Rank. cost	Assessments due before cut-off	+0.46	0.078
Pres. (22)	Drift	Withdrawn share among at-risk	−0.71	0.004
Pres. (22)	Drift	Share aged 35+	+0.62	0.023
Pres. (22)	Drift	At-risk rate	−0.61	0.023
Pres. (22)	Drift	Share without A-level	+0.58	0.029
Pres. (22)	Drift	Median clicks	−0.57	0.029
Pres. (22)	Drift	Assessment-structure distance	+0.53	0.042
Pres. (22)	Drift	VLE activity types	−0.48	0.063
Pres. (22)	Drift	Assessments due before cut-off	−0.45	0.082

V. DISCUSSION

A. What transfers and what does not

Our results draw a sharper picture of model portability than an accuracy gap. The ordering of students by risk is largely shared across modules: early scores, recent activity and the number of submissions predict risk everywhere, as local models confirm. What is specific to a course is the scale on which these signals are expressed: how many clicks a normal week requires, when the first assessment is due, whether early assignments are graded. A model that learns absolute thresholds carries the scale of its training courses into the new one. The parallel with clinical prediction models, where discrimination travels better than calibration [15], [16], is close, and so are the remedies.

B. Implications for deployment in a new context

For an institution starting an early-warning system without labelled history, three practical recommendations follow. First, express engagement and early-assessment features relative to the cohort at the prediction date. This needs no outcome data, only a cohort observed up to the same day, which online programmes have. Second, monitor the alert volume against expectations from the first weeks: a flag rate far from the expected risk is a warning sign of threshold drift, not a sign of an at-risk cohort. Third, once a first cohort is labelled, recalibrate; around a hundred labelled students sufficed here. The same reasoning applies across countries, where VLE usage and assessment practices may differ more than between modules of one university.

C. Implications for evaluation practice

Four lessons generalize beyond OULAD. Early-prediction studies should restrict the population to students still enrolled at the prediction date and check leakage by invariance, not only by excluding outcome variables. Comparisons between protocols must use the same aggregation. Threshold-based metrics should be reported with the alert rate, since recall alone rewards over-flagging. Fairness audits that pool contexts should be complemented by within-context audits; otherwise a calibration problem can read as discrimination and, conversely, a real within-context bias can be masked by composition.

D. Limitations

The study uses one institution and seven modules; module-level correlations have little power, and presentation-level ones are not independent. We use a single model family and a fixed threshold; gradient boosting or sequential models

TABLE V

FLAG-RATE DRIFT (B) − (A) AND PRECISION OF THE TRANSFERRED MODEL UNDER THE TWO REMEDIES (25% CUT-OFF). PLATT-k: PLATT RECALIBRATION ON $k = 100$ RANDOMLY DRAWN LABELLED STUDENTS OF THE TARGET MODULE (MEAN OF 50 DRAWS, EVALUATED ON THE REMAINING STUDENTS). PLATT-PILOT: RECALIBRATION ON THE FIRST PRESENTATION OF THE MODULE, EVALUATED ON THE LATER ONES (DRIFT MEASURED AGAINST (A) ON THE SAME STUDENTS). RELATIVE: CONTEXT-RELATIVE FEATURES, NO LABELS.

	Flag-rate drift				Precision of (B)			
Module	Raw	Platt-k	Platt-pilot	Relative	Raw	Platt-k	Platt-pilot	Relative
AAA	+0.09	+0.07	+0.09	+0.03	0.38	0.38	0.38	0.46
BBB	+0.19	+0.06	+0.15	+0.04	0.54	0.60	0.53	0.67
CCC	−0.04	+0.03	−0.11	+0.03	0.78	0.74	0.85	0.73
DDD	−0.05	+0.03	+0.04	−0.02	0.79	0.75	0.73	0.81
EEE	+0.07	+0.10	+0.14	+0.01	0.69	0.66	0.62	0.72
FFF	−0.02	+0.08	+0.07	−0.01	0.76	0.69	0.70	0.76
GGG	+0.51	−0.04	+0.08	−0.01	0.38	0.55	0.49	0.64
Mean $\lvert\cdot\rvert$ / mean	0.14	0.06	0.10	0.02	0.62	0.62	0.61	0.68

TABLE VI

BETWEEN-GROUP GAPS IN ERROR RATES AT THE 25% CUT-OFF (THRESHOLD 0.5), WITH 95% CONFIDENCE INTERVALS FROM A STUDENT-LEVEL CLUSTER BOOTSTRAP (1,000 RESAMPLES). TPR: RECALL; FPR: FALSE-POSITIVE RATE; PPV: PRECISION; FLAG: SHARE OF STUDENTS FLAGGED.

Contrast	Configuration	TPR gap	FPR gap	PPV gap	Flag-rate gap
Female − male	(A) absolute	−0.080 [−0.096, −0.061]	−0.007 [−0.019, +0.005]	−0.042 [−0.058, −0.025]	−0.050 [−0.062, −0.038]
	(B) absolute	+0.014 [−0.003, +0.031]	+0.188 [+0.174, +0.201]	−0.168 [−0.184, −0.151]	+0.105 [+0.093, +0.117]
	(B) relative	−0.062 [−0.080, −0.044]	+0.016 [+0.004, +0.027]	−0.061 [−0.077, −0.043]	−0.029 [−0.042, −0.018]
IMD Q1 − Q5	(A) absolute	+0.135 [+0.104, +0.165]	+0.148 [+0.127, +0.169]	+0.044 [+0.013, +0.077]	+0.224 [+0.205, +0.243]
	(B) absolute	+0.143 [+0.113, +0.173]	+0.163 [+0.139, +0.189]	+0.102 [+0.073, +0.133]	+0.223 [+0.203, +0.245]
	(B) relative	+0.125 [+0.092, +0.157]	+0.113 [+0.092, +0.133]	+0.062 [+0.032, +0.095]	+0.203 [+0.182, +0.224]
Disability yes − no	(A) absolute	+0.095 [+0.067, +0.121]	+0.102 [+0.076, +0.128]	+0.020 [−0.006, +0.047]	+0.146 [+0.124, +0.167]
	(B) absolute	+0.060 [+0.034, +0.085]	+0.098 [+0.069, +0.126]	+0.047 [+0.022, +0.076]	+0.119 [+0.097, +0.140]
	(B) relative	+0.080 [+0.052, +0.107]	+0.064 [+0.039, +0.090]	+0.041 [+0.015, +0.068]	+0.120 [+0.097, +0.141]

TABLE VII

ROBUSTNESS TO THE TARGET DEFINITION (25% CUT-OFF, STUDENTS STILL ENROLLED; MEANS OVER MODULES). W OR F: WITHDRAWN OR FAIL; W ONLY: WITHDRAWN ONLY. ABS./REL.: ABSOLUTE OR RELATIVE FEATURES. COST: AUC (A) − AUC (B); DRIFT: MEAN ABSOLUTE FLAG-RATE DRIFT.

Target	Feat.	Risk (%)	AUC (A)	AUC (B)	Cost	Drift	GGG
W or F	abs.	38.7	0.800	0.787	+0.013	0.137	+0.51
W or F	rel.	38.7	0.816	0.809	+0.008	0.021	−0.01
W only	abs.	13.7	0.651	0.630	+0.021	0.052	+0.18
W only	rel.	13.7	0.675	0.668	+0.006	0.028	+0.03

may shift absolute numbers, although the mechanism we identify concerns features rather than learners. SHAP values are estimated from a subsample of trees and 400 students per module. Relative ranks remove between-cohort differences in the level of engagement, so a model using them cannot detect a whole cohort that is collectively disengaged; in very small cohorts, ranks are also noisy. IMD gaps remain and call for policy choices on how alerts are used. Finally, the transfer we study is between modules of one university; transfer across institutions and countries, which motivates this work, remains to be tested.

E. Future work

We will replay the same protocol on Moodle data from a fully online Senegalese university, where the held-out context becomes a degree programme and then a country, and compare sequential models with the aggregated features used here.

VI. CONCLUSION

Measured honestly—early, on students still enrolled, and with comparable aggregation—the cost of moving an at-risk model to an unseen course is small for ranking and large for thresholds. The threshold fails because engagement and early scores are read on absolute scales that depend on course design; the same features matter everywhere. Expressing them relative to the cohort removes the drift without labels, improves ranking on unseen courses, and removes a gender disparity created by pooling contexts. Early-warning systems meant to travel should be evaluated on unseen contexts, report alert rates alongside accuracy, and audit fairness within contexts.

DATA AND CODE AVAILABILITY

OULAD is available under a CC-BY 4.0 licence from the Open University (https://analyse.kmi.open.ac.uk/open_dataset). The code that reproduces every figure, table and number of this article is available from the authors and will be released in a public repository upon publication.

REFERENCES

[1] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open university learning analytics dataset," *Scientific Data*, vol. 4, p. 170171, 2017.

[2] L. Jin, Y. Wang, H. Song, and H. J. So, "Predictive modelling with the Open University Learning Analytics Dataset (OULAD): A systematic literature review," in *Artificial Intelligence in Education. Posters and Late Breaking Results, Workshops and Tutorials, Industry and Innovation Tracks, Practitioners, Doctoral Consortium and Blue Sky (AIED 2024)*. Springer, 2024, pp. 477–484.

[3] A. Elsayed, Y. Wardat, F. Kamalov, and H. Sulieman, "Toward sustainable and equitable AI in education through a regional fairness audit of dropout prediction using the OULAD dataset," *Sustainability*, vol. 18, no. 18, p. 9440, 2026.

[4] C. Malkani, N. Mittal, and S. Som, "A hybrid deep Kronecker and BN-eLU-AlexNet framework for early prediction of learner outcomes in e-learning platforms," *Frontiers in Computer Science*, vol. 8, p. 1889642, 2026.

[5] S. Wu, J. Duan, and M. Luo, "Calibrated early-warning models with fairness auditing and selective prediction for course withdrawal risk: Evidence from OULAD," *PLOS ONE*, vol. 21, no. 7, p. e0352867, 2026.

[6] D. Gašević, S. Dawson, T. Rogers, and D. Gašević, "Learning analytics should not promote one size fits all: The effects of instructional conditions in predicting academic success," *The Internet and Higher Education*, vol. 28, pp. 68–84, 2016.

[7] R. Conijn, C. Snijders, A. Kleingeld, and U. Matzat, "Predicting student performance from LMS data: A comparison of 17 blended courses using Moodle LMS," *IEEE Transactions on Learning Technologies*, vol. 10, no. 1, pp. 17–29, 2017.

[8] J. López-Zambrano, J. A. Lara, and C. Romero, "Towards portability of models for predicting students' final performance in university courses starting from Moodle logs," *Applied Sciences*, vol. 10, no. 1, p. 354, 2020.

[9] M. Adnan *et al.*, "Predicting at-risk students at different percentages of course length for early intervention using machine learning models," *IEEE Access*, vol. 9, pp. 7519–7539, 2021.

[10] M. Hlosta, Z. Zdrahal, and J. Zendulka, "Ouroboros: Early identification of at-risk students without models based on legacy data," in *Proc. 7th Int. Learning Analytics and Knowledge Conf. (LAK'17)*. ACM, 2017, pp. 6–15.

[11] S. M. Jayaprakash, E. W. Moody, E. J. M. Lauría, J. R. Regan, and J. D. Baron, "Early alert of academically at-risk students: An open source analytics initiative," *Journal of Learning Analytics*, vol. 1, no. 1, pp. 6–47, 2014.

[12] M. Riestra-González, M. d. P. Paule-Ruíz, and F. Ortín, "Massive LMS log data analysis for the early prediction of course-agnostic student performance," *Computers & Education*, vol. 163, p. 104108, 2021.

[13] S. Boyer and K. Veeramachaneni, "Transfer learning for predictive models in massive open online courses," in *Artificial Intelligence in Education (AIED 2015)*, ser. Lecture Notes in Computer Science, vol. 9112. Springer, 2015, pp. 54–63.

[14] J. G. Moreno-Torres, T. Raeder, R. Alaiz-Rodríguez, N. V. Chawla, and F. Herrera, "A unifying view on dataset shift in classification," *Pattern Recognition*, vol. 45, no. 1, pp. 521–530, 2012.

[15] K. J. M. Janssen, K. G. M. Moons, C. J. Kalkman, D. E. Grobbee, and Y. Vergouwe, "Updating methods improved the performance of a clinical prediction model in new patients," *Journal of Clinical Epidemiology*, vol. 61, no. 1, pp. 76–86, 2008.

[16] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, "Calibration: the Achilles heel of predictive analytics," *BMC Medicine*, vol. 17, p. 230, 2019.

[17] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*, A. J. Smola, P. Bartlett, B. Schölkopf, and D. Schuurmans, Eds. MIT Press, 1999, pp. 61–74.

[18] J. Gardner, C. Brooks, and R. Baker, "Evaluating the fairness of predictive student models through slicing analysis," in *Proc. 9th Int. Conf. on Learning Analytics & Knowledge (LAK'19)*. ACM, 2019, pp. 225–234.

[19] R. S. Baker and A. Hawn, "Algorithmic bias in education," *International Journal of Artificial Intelligence in Education*, vol. 32, no. 4, pp. 1052–1092, 2022.

[20] R. A. Opoku, B. Pei, and W. Xing, "Unveiling accuracy-fairness trade-offs: Investigating machine learning models in student performance prediction," *Journal of Learning Analytics*, vol. 12, no. 2, pp. 125–139, 2025.

[21] A. Chouldechova, "Fair prediction with disparate impact: A study of bias in recidivism prediction instruments," *Big Data*, vol. 5, no. 2, pp. 153–163, 2017.

[22] J. Kleinberg, S. Mullainathan, and M. Raghavan, "Inherent trade-offs in the fair determination of risk scores," in *8th Innovations in Theoretical Computer Science Conf. (ITCS 2017)*, ser. LIPIcs, vol. 67, 2017, pp. 43:1–43:23.

[23] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," in *Advances in Neural Information Processing Systems 29 (NeurIPS 2016)*, 2016, pp. 3315–3323.

[24] E. H. Simpson, "The interpretation of interaction in contingency tables," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 13, no. 2, pp. 238–241, 1951.

[25] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 15:1–15:21, 2012.

[26] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[27] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[28] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: A practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.

[29] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017, pp. 4765–4774.

[30] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.

[31] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman & Hall, 1993.